

Network Science

Hubs

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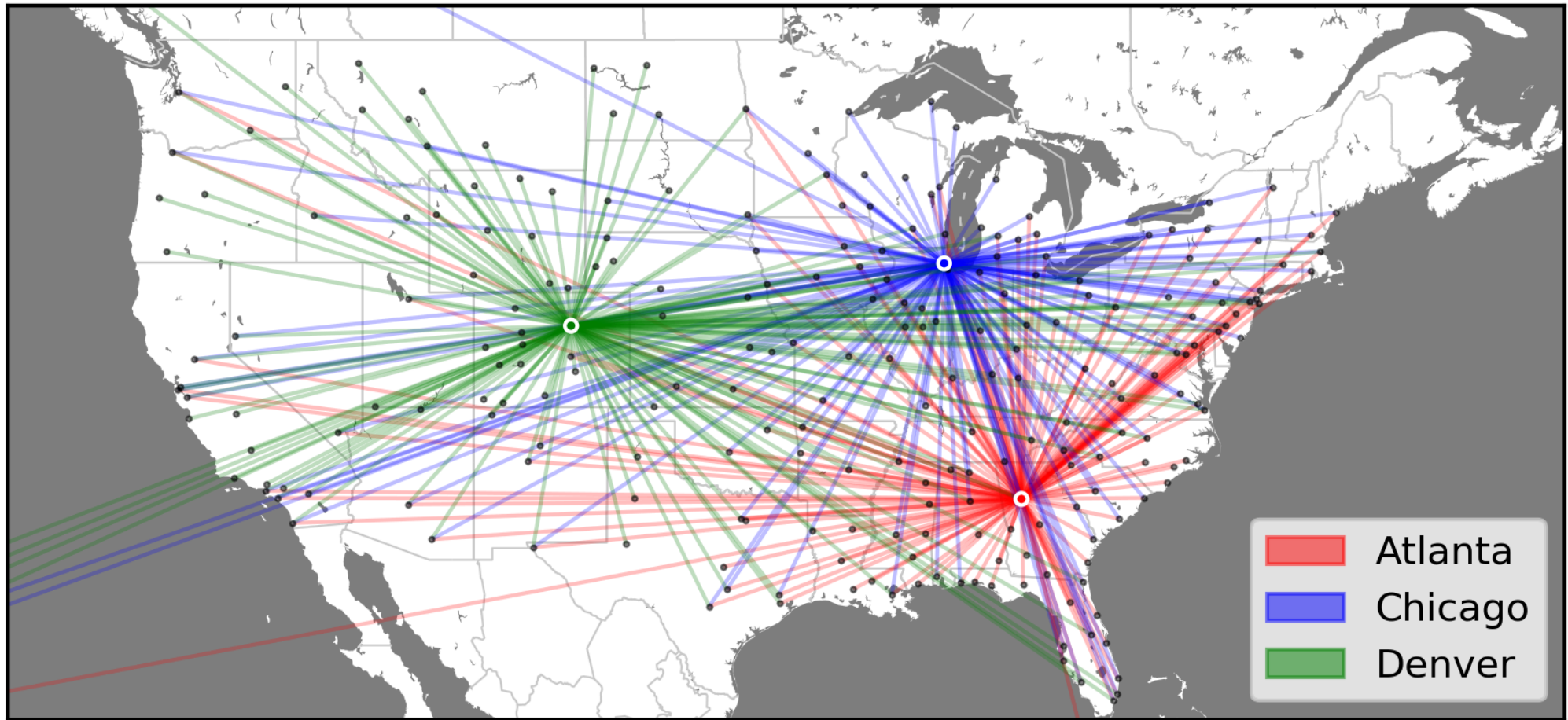
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Chapter 3: Hubs

- Centrality measures
- Centrality distributions
- The friendship paradox
- Ultra-small worlds
- Robustness
- Core decomposition

Real networks are heterogeneous



Some nodes (and links) are much more important (**central**) than others!

Centrality measures

- **Centrality** : measure of importance of a node
- **Measures** :
 - Degree
 - Closeness
 - Betweenness

! Important

There are many more centrality measures. For a survey, see [1] and [2]. I like [2] because it also mentions centrality measures for multilayer and temporal networks which do require particular treatment.

Degree

Reminder: The degree k_i of a node i is the number of its neighbours

High-degree nodes are called **hubs**

The *average degree* of an (*undirected*) network is given by:

$$\langle k \rangle = \frac{\sum_i k_i}{N} = \frac{2L}{N}$$

```
1 G.degree(2) # returns the degree of node 2
2 G.degree()  # dict with the degree of all nodes of G
```

Closeness

Idea: a node is the more central the *closer* it is to the other nodes, on average

$$g_i = \frac{1}{\sum_{j \neq i} \ell_{ij}}$$

where ℓ_{ij} is the distance between nodes i and j

```
1 nx.closeness centrality(G, node) # closeness centrality of node
```

Betweenness (1/2)

Idea: a node is the more central the *more often it is crossed by paths*

$$b_i = \sum_{h \neq j \neq i} \frac{\sigma_{hj}(i)}{\sigma_{hj}}$$

where

σ_{hj} is the number of shortest paths from h to j

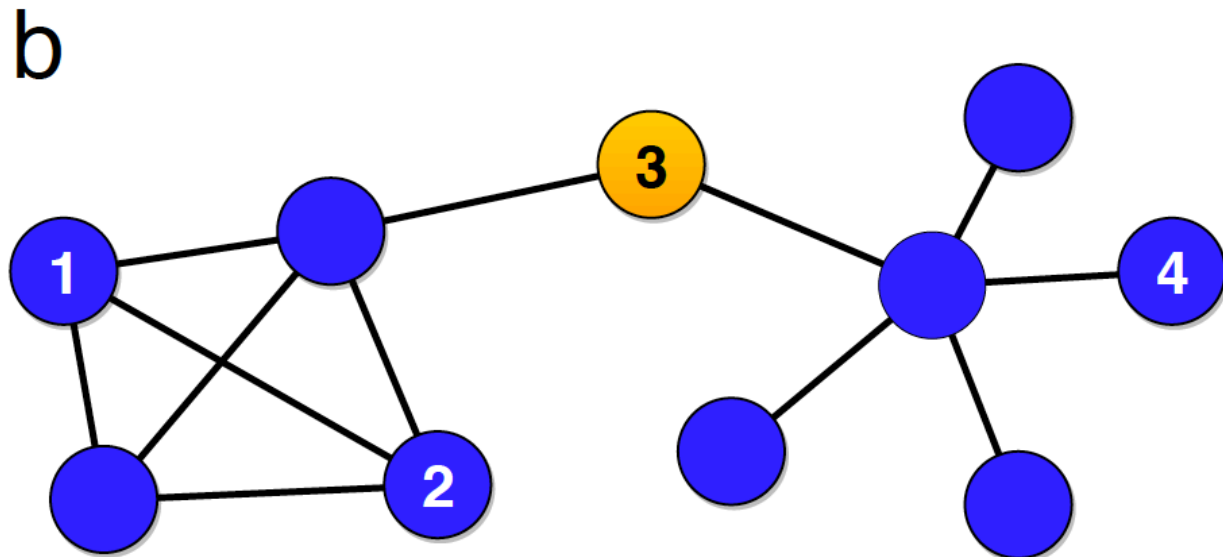
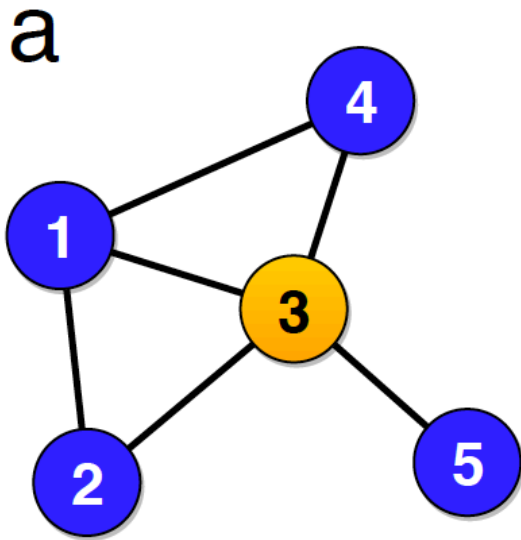
$\sigma_{hj}(i)$ is the number of shortest paths from h to j running through i

Note

Betweenness can be easily extended to links. **Link betweenness** : fraction of shortest paths among all possible node pairs that pass through the link

Betweenness (2/2)

Hubs usually have high betweenness, but there can be nodes with high betweenness **that are not hubs**



3: 0.58

3: 0.56

1: 0.08

1: 0.10

2: 0.00

2: 0.10

4: 0.00

5: 0.00

5: 0.00

6: 0.50

7: 0.72

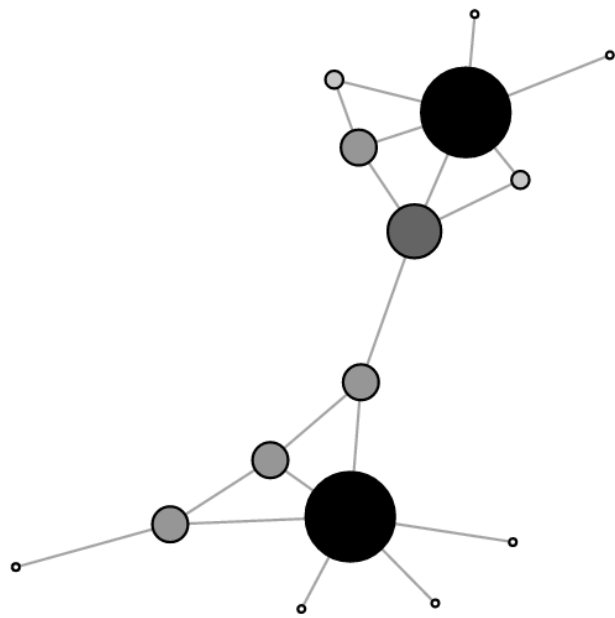
4: 0.00

8: 0.00

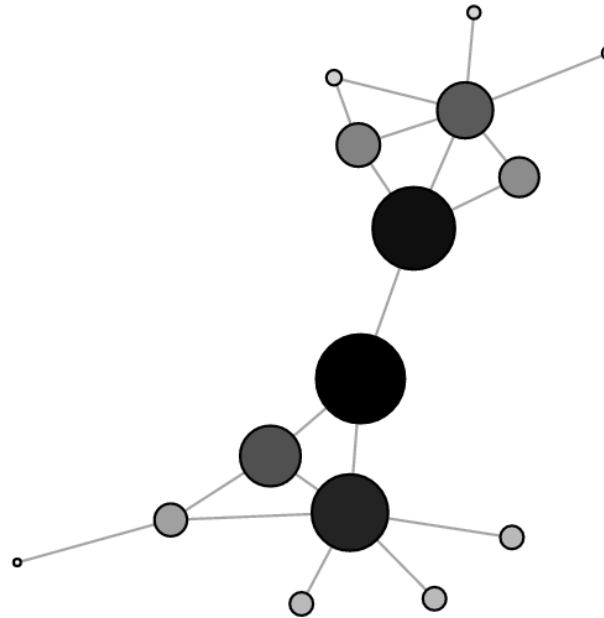
9: 0.00

10: 0.00

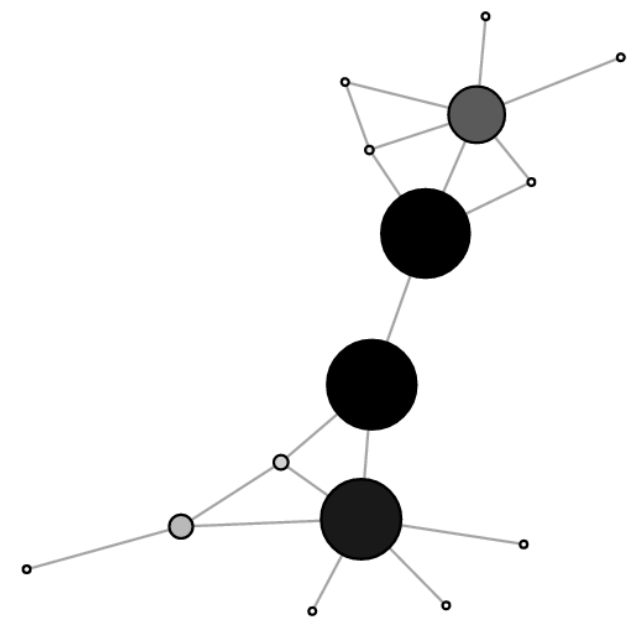
A quick visual depiction of differences



(a) Degree



(c) Closeness centrality



(d) Betweenness centrality

Reproduced from [2]

Centrality distributions (1/3)

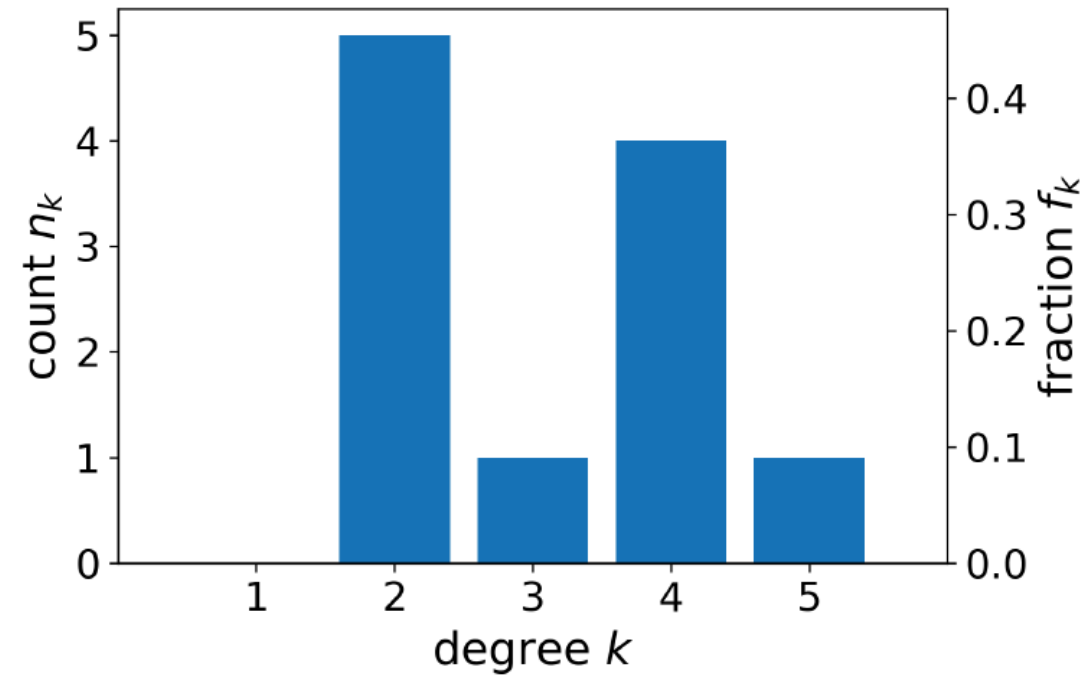
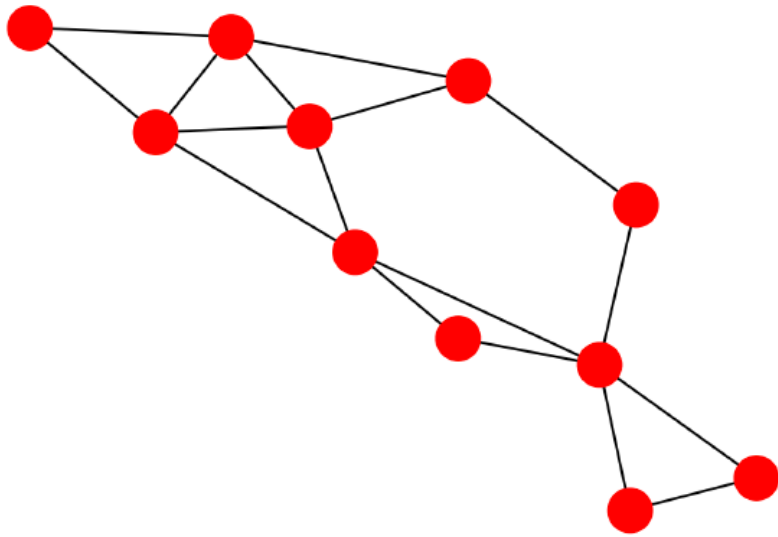
On small networks it makes sense to ask which nodes or links are most important

On large networks **it does not**

Solution: statistical approach

Instead of focusing on individual nodes and links, we consider **classes** of nodes and links with similar properties

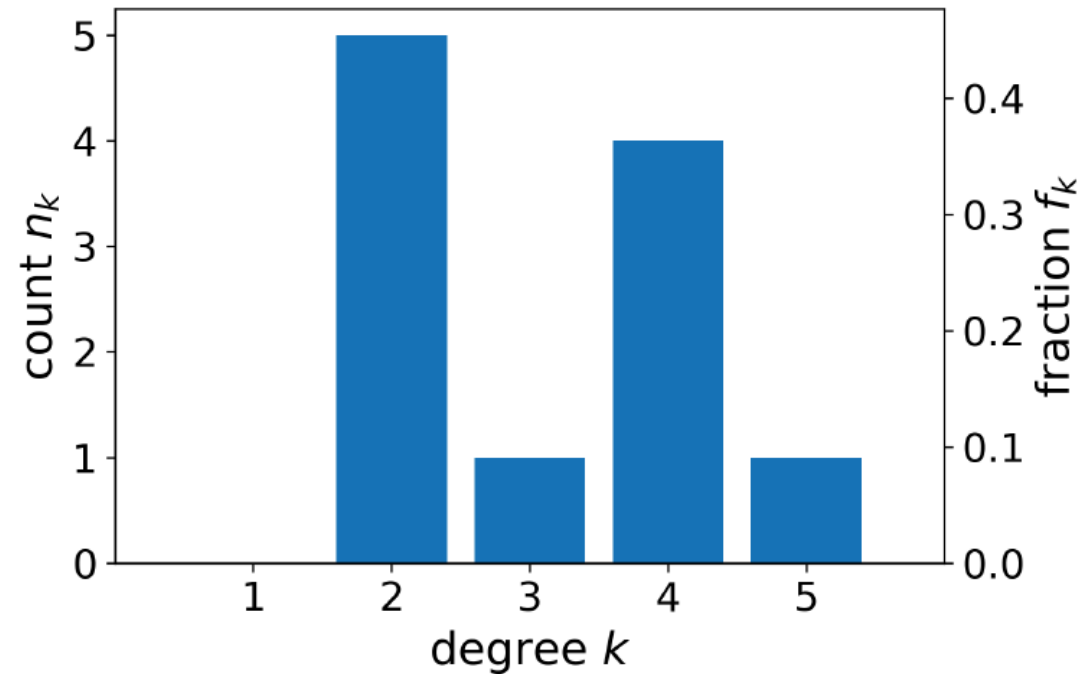
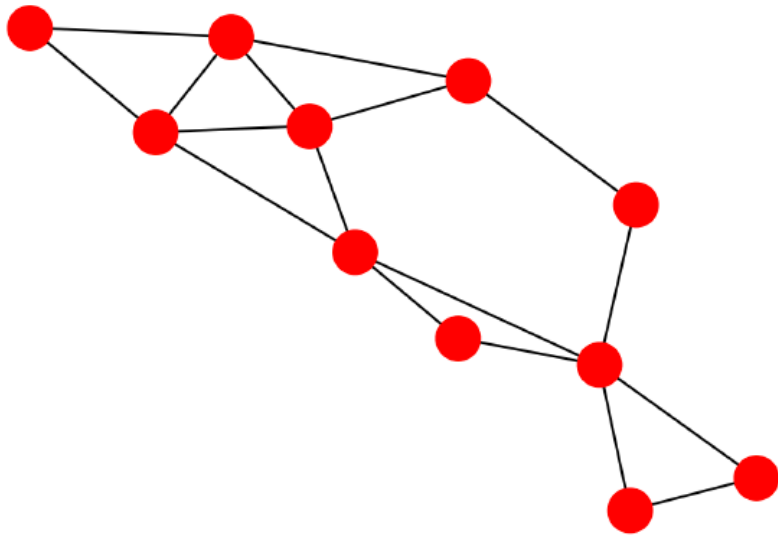
Centrality distributions (2/3)



n_k is the number of nodes with degree k

$f_k = \frac{n_k}{N}$ is the frequency of degree k

Centrality distributions (3/3)



When $N \rightarrow \infty$, f_k becomes the **probability** p_k of having degree k

p_k versus k is the **probability distribution** of node degree

Cumulative distributions

If the variable is *not integer* (e.g., betweenness), the range of the variable is divided into intervals (bins) and we count how many values fall in each interval

Cumulative distribution $P(x)$: probability that the variable takes values *larger* than x as a function of x

It is complementary to the probability distribution and often used when the range of variability is very broad (as is often the case with centrality measures in real-world networks).

How to compute it? by summing the frequencies of the variable inside the intervals to the right of x

$$P(x) = \sum_{v \geq x} f_v$$

Logarithmic scale

Question: how to plot a probability distribution if the variable spans a large range of values, from small to (very) large?

Answer: use the **logarithmic** scale

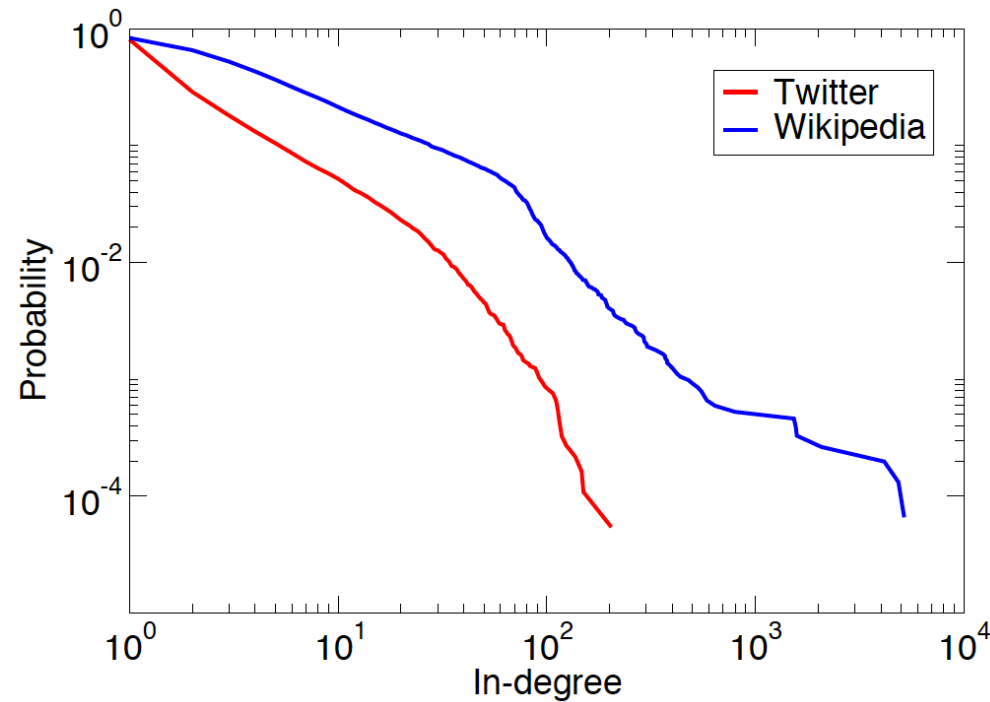
How to do it? report the logarithms of the values on the x- and y-axes

$$\log_{10}(10) = 1$$

$$\log_{10} 1000 = \log_{10} 10^3 = 3$$

$$\log_{10} 1000000 = \log_{10} 10^6 = 6$$

Degree distributions (1/2)



Heavy-tail distributions: the variable goes from small to large values

i Note

A bit more formally, we say a distribution is heavy-tailed if the frequency of large values is much greater than expected in an exponential distribution, where it is vanishingly small and contributes little to mean and variance.

Degree distributions (2/2)

The *heterogeneity* parameter κ says how broad the distribution is:

$$\kappa = \frac{\langle k^2 \rangle}{\langle k \rangle^2}$$

$$\langle k \rangle = \frac{\sum_i k_i}{N} = \frac{2L}{N}; \quad \langle k^2 \rangle = \frac{\sum_i k_i^2}{N}$$

If most degrees have the same value, say k_0 :

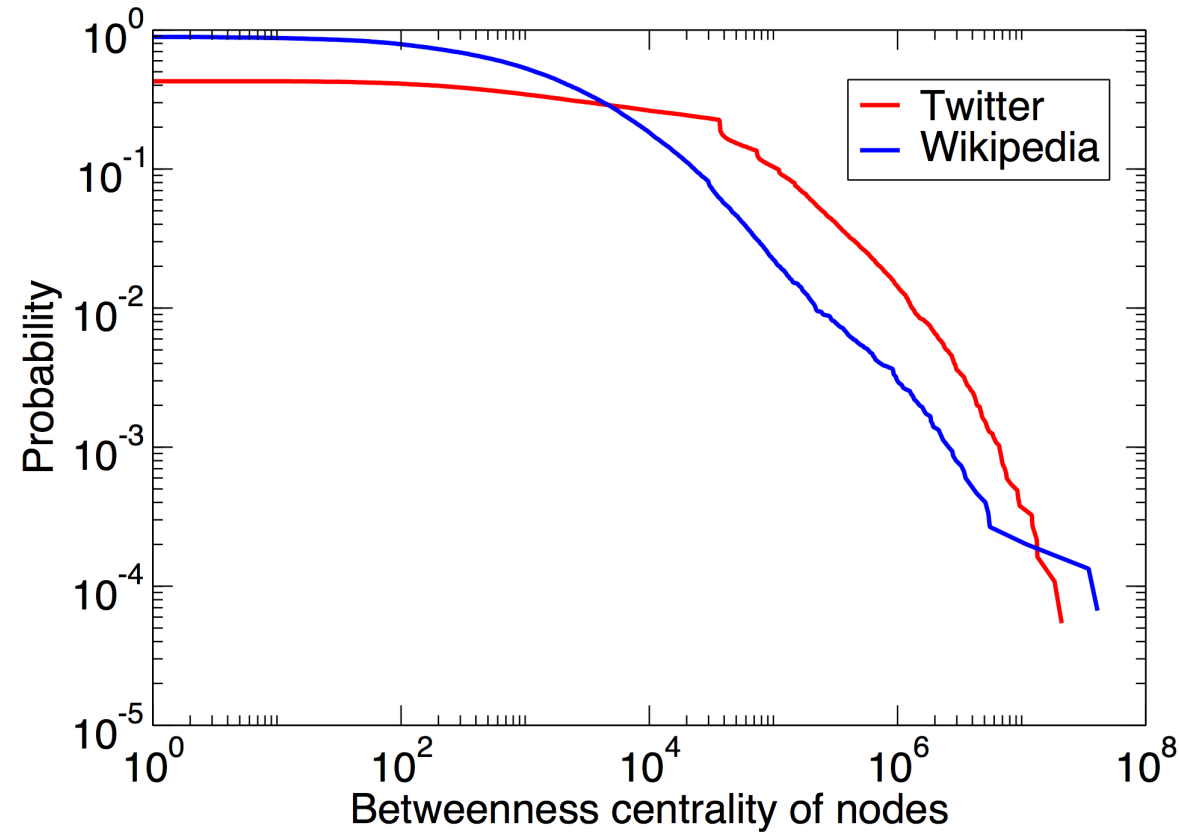
$$\langle k \rangle \approx k_0, \quad \langle k^2 \rangle \approx k_0^2 \implies \kappa \approx 1$$

If the distribution is very heterogeneous: $\kappa \gg 1$

Degree centrality

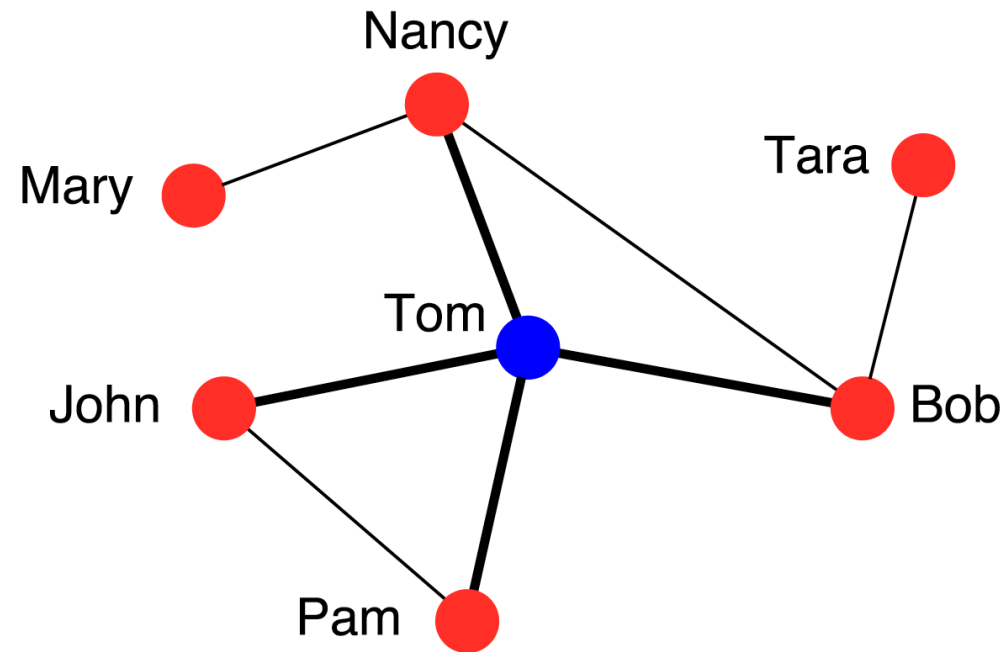
Network	Nodes (N)	Links (L)	Average degree ($\langle k \rangle$)	Maximum degree (k_{max})	Heterogeneity parameter (κ)
Facebook Northwestern Univ.	10,567	488,337	92.4	2,105	1.8
IMDB movies and stars	563,443	921,160	3.3	800	5.4
IMDB co-stars	252,999	1,015,187	8.0	456	4.6
Twitter US politics	18,470	48,365	2.6	204	8.3
Enron Email	36,692	367,662	10.0	1,383	14.0
Wikipedia math	15,220	194,103	12.8	5,171	38.2
Internet routers	190,914	607,610	6.4	1,071	6.0
US air transportation	546	2,781	10.2	153	5.3
World air transportation	3,179	18,617	11.7	246	5.5
Yeast protein interactions	1,870	2,277	2.4	56	2.7
C. elegans brain	297	2,345	7.9	134	2.7
Everglades ecological food web	69	916	13.3	63	2.2

Betweenness distributions



Heavy-tail distributions: the variable goes from small to large values

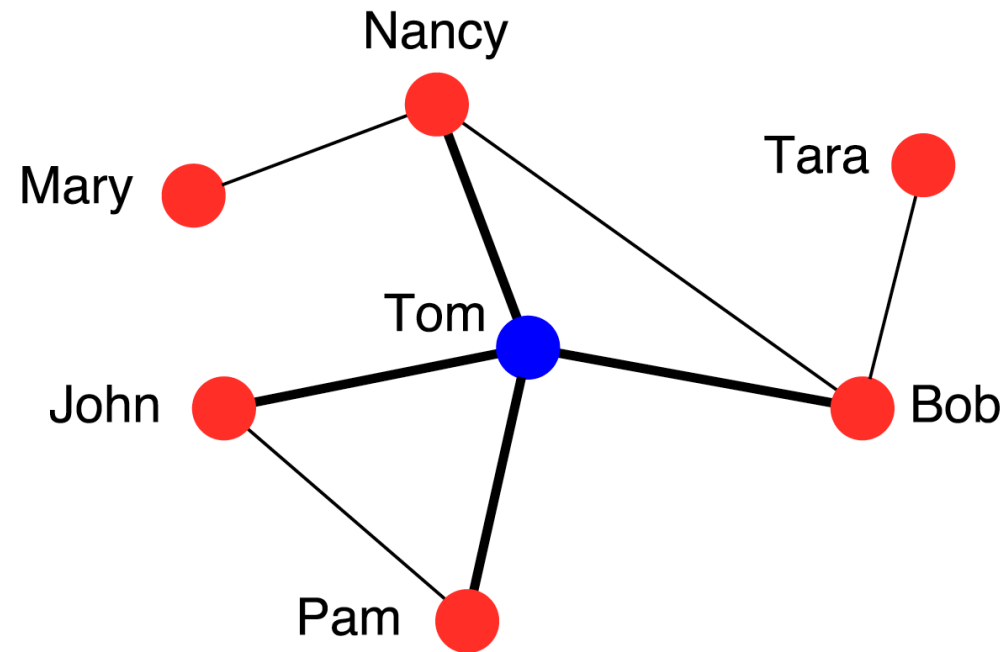
Friendship paradox (1/4)



Choosing *nodes at random* , Tom has **the same chance** to be picked as anyone else

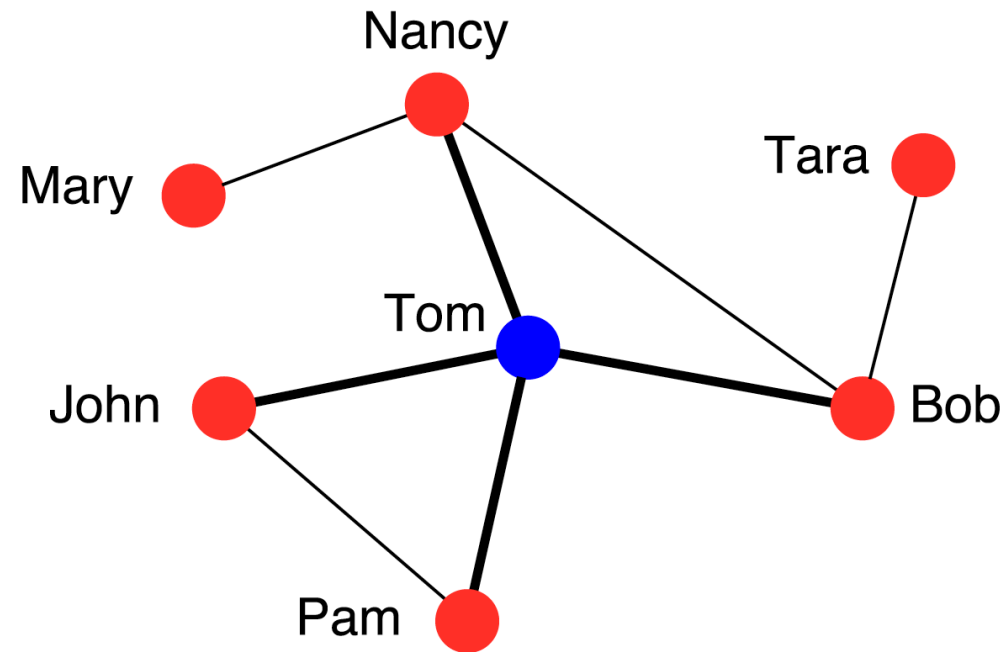
Choosing *links at random* , Tom has a **higher chance** to be picked

Friendship paradox (2/4)



By following links, **the chance to hit a hub increases**

Friendship paradox (3/4)



Average degree of a node = **2.29**

Average degree of the neighbours of a node = **2.83 > 2.29**

Our friends have more friends than we do, on average (**friendship paradox**)

Friendship paradox (4/4)

Question: Where does the friendship paradox come from?

Answer:

1. By averaging the degree of the nodes, we pick them at random
2. By averaging the degree of the neighbours, we choose them by following links and then nodes with degree k will be counted k times, which inflates the average

The more hubs, the stronger the effect (see lab class exercise!)

Ultra-small worlds (1/2)

In real networks, many shortest paths go through hubs

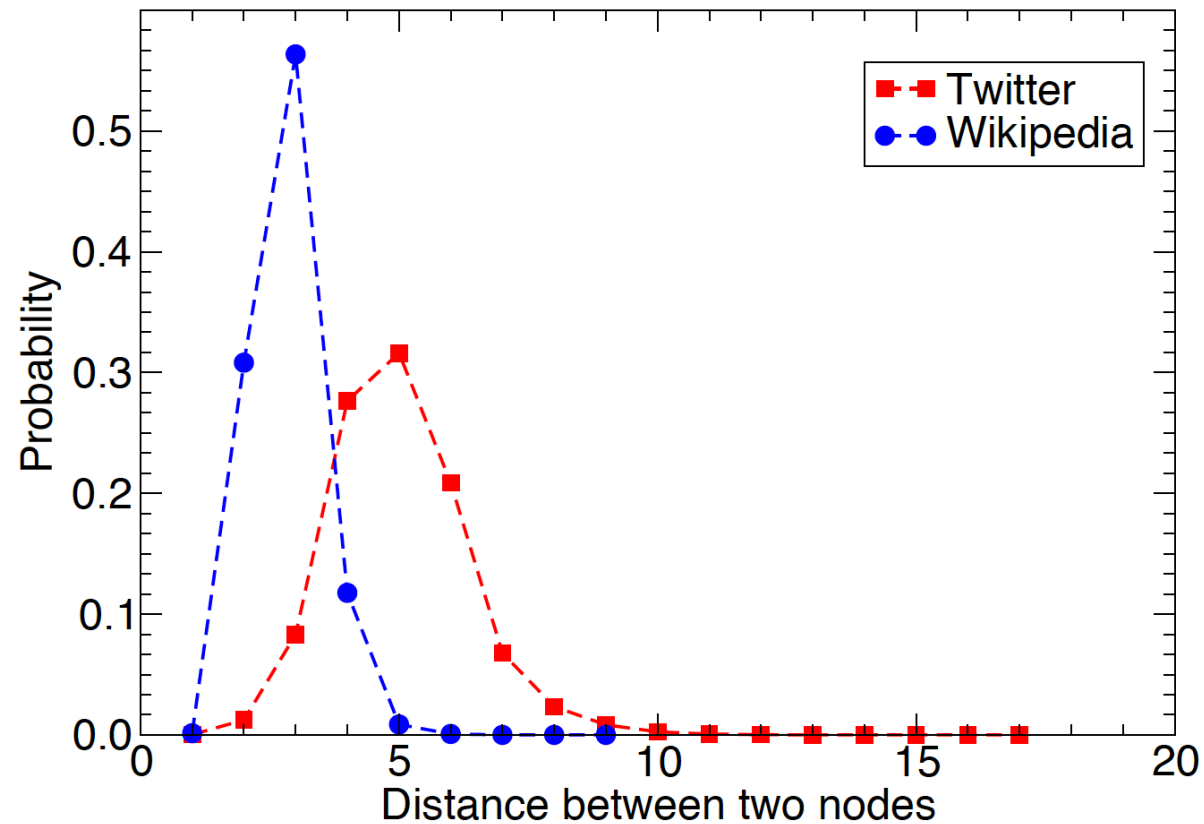
Example: air transportation

There may be no routes between airport A and B (if they are small), but it may be possible to go from A to B via a hub airport C

The *small-world* property is typical of most networks of interest: if the network has hubs, paths are ultra-short (**ultra-small world**)

Ultra-small worlds (1/2)

Shortest-path length distribution



Robustness (1/5)

A system is **robust** if the failure of some of its components does not affect its function

Question: how can we define the robustness of a network?

Answer: we remove nodes and/or links and see what happens to its structure

Key point: *connectedness*

If the Internet were not connected, it would be impossible to transmit signals (e.g., emails) between routers in different components

Warning

Note that we *defined* robustness in terms of *function*, but we *evaluated* impact of removal on *structure*. What are the assumptions being made here? Is function always sensitive to changes in structure?

Robustness (2/5)

Robustness test: checking how the connectedness of the network is affected as more and more nodes are removed

How to do it: plot the relative size S of the largest connected component as a function of the fraction of removed nodes

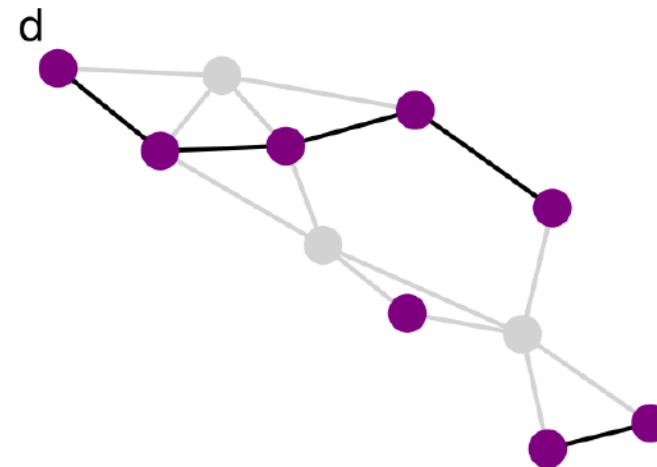
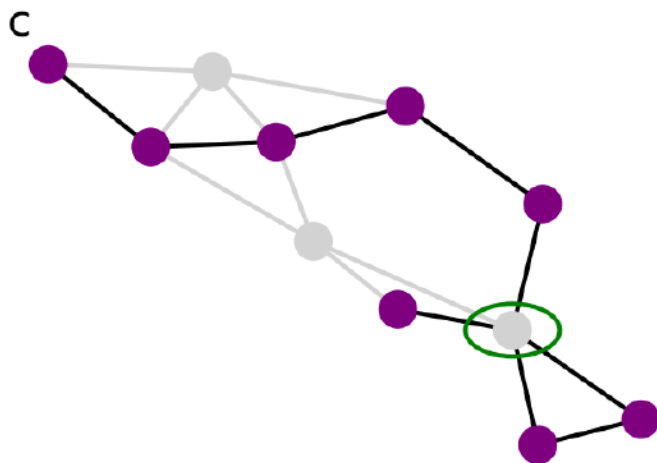
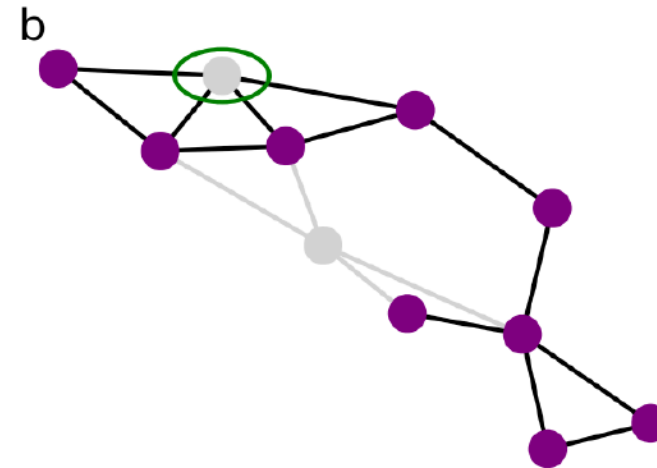
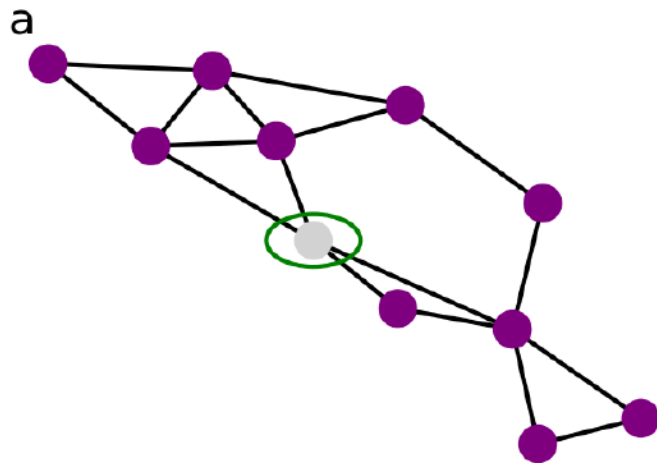
We suppose that the network is initially connected: there is only one component and $S = 1$

As more and more nodes (and their links) are removed, the network is progressively broken up into components and S goes down

Note

This is the subject of an exercise in a lab-class

Robustness (3/5)

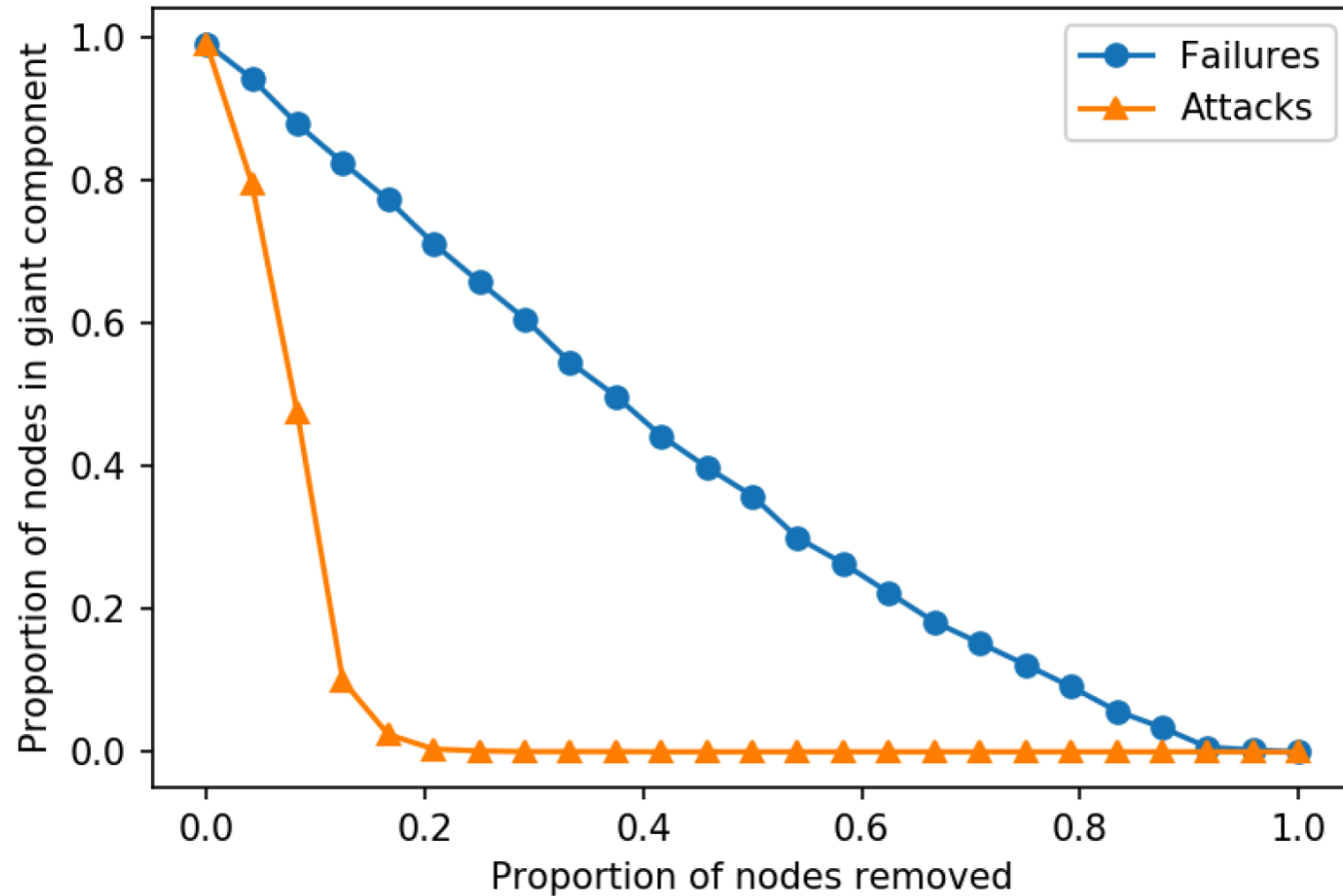


Robustness (4/5)

Two strategies:

1. **Random failures**: nodes break down randomly, so they are all chosen with the **same probability**. Concretely, we remove a fraction f of nodes, chosen at random
2. **Attacks**: hubs are deliberately targeted — the larger the **degree**, the higher the probability of removing the node. Concretely, remove the fraction f of nodes with largest degree, from the one with largest degree downwards

Robustness (5/5)



Conclusion: real networks (here, OpenFlights World network) are robust against random failures but fragile against targeted attacks!

Core decomposition (1/3)

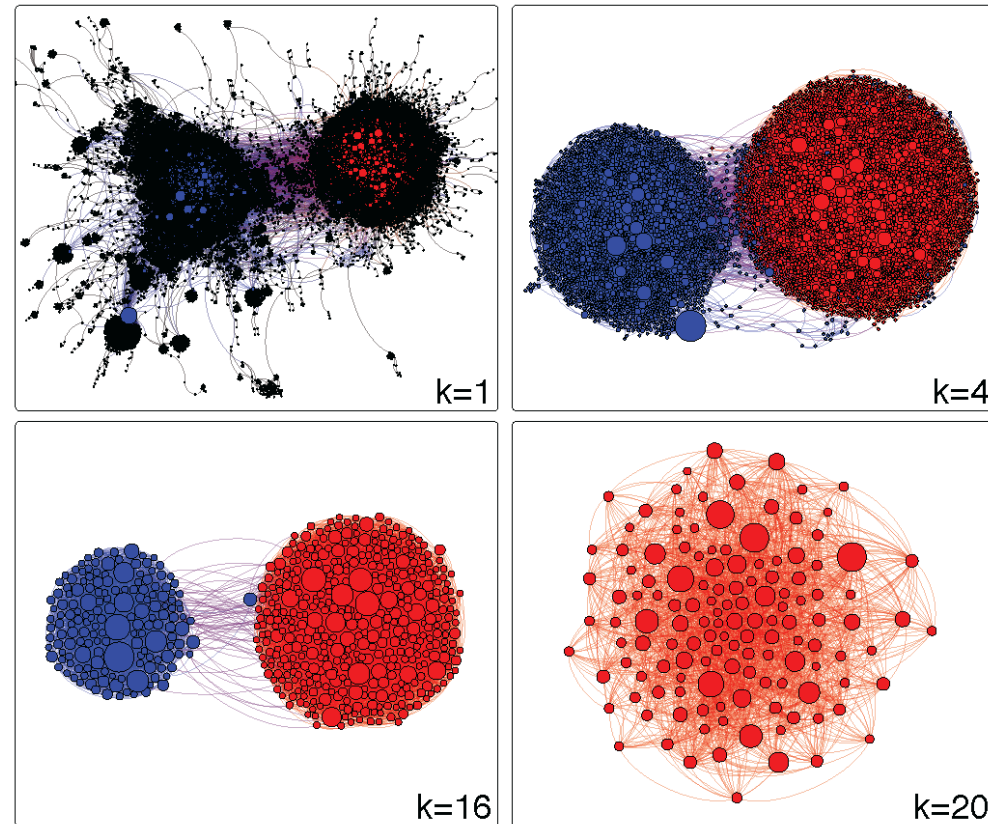
Core: dense part of the network, with high-degree nodes

Core decomposition: procedure to identify denser and denser cores, by removing nodes of progressively higher degree. If we remove all nodes with degree $k - 1$ or lower, the remaining portion of the network is called k -core

k -core decomposition procedure:

0. Start with $k = 0$
1. **Recursively** remove all nodes with degree k until none are left
2. The set of removed nodes is the k -th shell, while the remaining ones form the $(k+1)$ -core
3. If there is no node left, terminate. Otherwise, increment k by one and repeat from step 1.

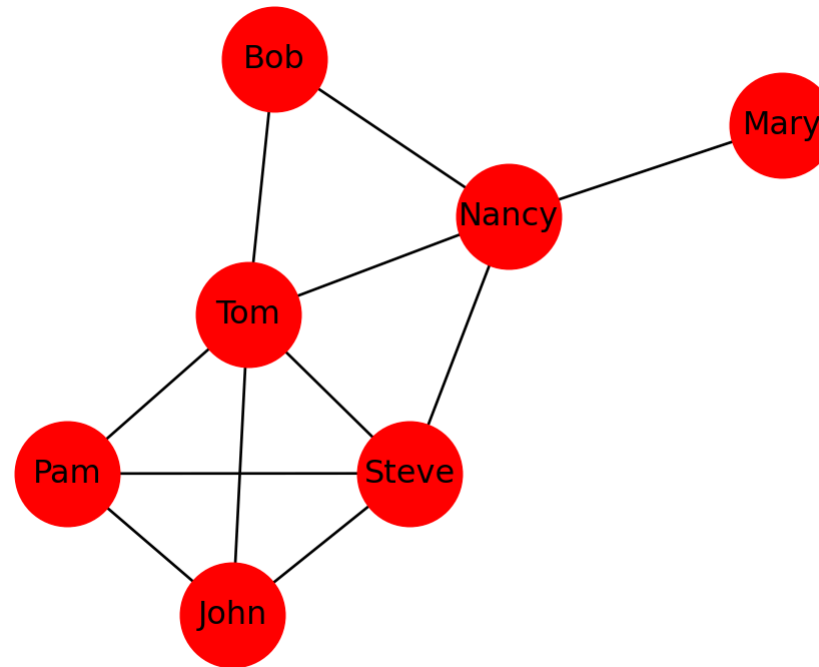
Core decomposition (2/3)



Twitter political retweet network

Core decomposition (3/3)

Core decomposition helps to visualize large networks, by pruning low-degree nodes and showing only the densest parts



```
1 nx.core_number(G)
2 nx.k_core(G,2)
3 nx.k_shell(G,2)
4 nx.k_core(G)
```

```
{'Mary': 1, 'Nancy': 2, 'Tom': 3, 'Bob': 2, 'Pam': 3, 'John': 3, 'Steve': 3}
Graph with 6 nodes and 10 edges
Graph with 2 nodes and 1 edges
Graph with 4 nodes and 6 edges
```

References

1. A. Saxena & S. Iyengar, Centrality Measures in Complex Networks: A Survey. (2020).
<https://doi.org/10.48550/arXiv.2011.07190>.
2. F. A. Rodrigues, Network centrality: An introduction. (2019).
<https://doi.org/10.48550/arXiv.1901.07901>.